

CS (Diploma) — Viruses, System Analysis & Design, and Internet Technology

Covers three separate syllabus units that are grouped here because they are all **vocabulary-heavy, low-maths, high-yield** topics — the kind that hand you cheap marks if you know the standard phrasing, and cost you nothing to learn:

- **Paper-I Unit 9 — Computer Viruses**
- **Paper-II Unit 4 — System Analysis & Design**
- **Paper-I Unit 6 — Internet Technology**

Two of the three are on **Paper-I** (viruses, internet) and one is on **Paper-II** (system analysis). All three are also heavily mined for MCQs — the real 2024 paper had ~15 questions drawn straight from these units.

Why this file matters

You are a B.Tech CS graduate. Your instinct will be to skip these units as "trivial" — and that instinct will cost you marks. Here is the honest situation:

1. **Viruses.** You know modern malware taxonomy (ransomware, rootkits, RATs, supply-chain attacks). The syllabus does **not** ask for that. It asks for a **1990s-era taxonomy** — "command-processor infection", "file-specific infection", "memory-resident infection" — phrases you have almost certainly never used. The examiner wants *their* categories back, in *their* words. Learn the vocabulary, not the concept (you already have the concept).
2. **System Analysis & Design.** This is management-information-systems vocabulary (Elias M. Awad / classic SAD textbooks), not software-engineering-as-you-were-taught. "Physical vs abstract system", "open vs closed system", "deterministic vs probabilistic", "TPS / MIS / DSS / ESS", "role of the system analyst", "factors causing failure" — these are *definitional* answers. A degree course teaches you to *build* software; this unit tests whether you can *classify systems* and *name SDLC phases* in the textbook's own terms.
3. **Internet Technology.** The connectivity types — **dial-up, leased line, ISDN, VSAT** — are dated and specific. You may never have configured a dial-up modem in your life. These are pure recall, and they generate easy single-fact MCQs (the 2024 paper had a full matching question on VSAT/dial-up/cable/leased).

Bottom line: none of this is hard. All of it is *unfamiliar phrasing of familiar ideas*.

The failure mode is over-confidence, not difficulty. Read for the vocabulary.

Tick-box checklist — you are done with this file when you can:

- [] List the **six virus types in the syllabus** (command-processor, boot-sector, executable-file, file-specific, memory-resident, macro) and give one distinguishing sentence for each — **without notes**.
- [] Distinguish **virus vs worm vs trojan vs ransomware** in a clean four-row table.
- [] Write the **symptoms of infection** and the **steps to deal with an infection** as numbered lists.
- [] Define **system**, list its **characteristics** and its **elements** (the "IPOFC" chain).
- [] Classify systems five ways: **physical/abstract, open/closed, deterministic/probabilistic, man-made/natural, permanent/temporary**.

- [] Name and one-line the four **information system** types: **TPS, MIS, DSS, ESS** (add OAS/KWS if asked).
- [] Recite the **SDLC phases in order** with the deliverable of each.
- [] State the **role of the system analyst** and the **factors that cause SDLC failure**.
- [] Contrast **intranet vs internet vs extranet**.
- [] Name the **internet services** (email, chat, BBS, video-conf, FTP, Telnet, WWW) and one line on each.
- [] Fill the **connectivity comparison table** (dial-up / leased / ISDN / VSAT / broadband) from memory.

How to use this file

Each topic follows the house format: **Concept** → **Key points** → **Worked example / illustration** → **Likely exam questions (5 / 10 / 15)** → **MCQ traps**. Numbers or facts I could not fully verify are flagged $\triangle$ **verify** — check them against a second source before quoting them as exact.

Mark scheme reminder. Descriptive Paper-I and Paper-II are each 100 marks / 3 hours:
 Section A = any **8 of 10 × 5**, Section B = any **3 of 5 × 10**, Section C = any **2 of 4 × 15**.
 MCQ = 100 × 2 = 200 with **−1/3 negative marking** per wrong answer. Definitions, tables and numbered lists score fastest in the descriptive papers — every topic below is built to deliver exactly those.

Topic checklist and where each unit sits

#	Topic	Paper	Depth	Best answer format
A	Computer viruses — definition, working	P-I §9	Medium	Definition + numbered life-cycle
B	Virus types (the syllabus's six)	P-I §9	High	Table, one row per type
C	Worms / trojans / ransomware (context)	P-I §9	Low	Four-row comparison table
D	Protection, recognising, dealing with infection	P-I §9	Medium	Three numbered lists
E	System concept, characteristics, elements	P-II §4	Medium	Definition + IPOFC diagram
F	Types of system	P-II §4	High	Five two-column tables
G	Information systems (TPS/MIS/DSS/ESS)	P-II §4	Medium	Pyramid + one-line each
H	SDLC phases	P-II §4	High	Numbered phases + deliverables
I	Role of system analyst; failure factors	P-II §4	Medium	Two numbered lists
J	Intranet / internet / extranet	P-I §6	Medium	Three-column table
K	Internet services	P-I §6	Medium	List, one line each
L	WWW, browsers, search engines	P-I §6	Medium	Definition + how-search-works
M	Connectivity (ISP, dial-up, leased, ISDN, VSAT)	P-I §6	High	Comparison table

PART 1 — COMPUTER VIRUSES (Paper-I, Unit 9)

A. What a virus is and how it works

Concept

A **computer virus** is a small program (a piece of malicious code) that **attaches itself to another program or file** and, when that host is run, **copies itself into other programs or files** — it *replicates*. That self-replication, riding on a host, is the defining property. The biological analogy is exact: a virus cannot live on its own; it hijacks a host cell (a legitimate program) and uses it to reproduce.

Two properties together define a virus:

1. **It replicates** (makes copies of itself), and
2. **It needs a host / a trigger** — it does not run by itself; it runs when the infected program runs, the infected document is opened, or the infected disk is booted.

A program that replicates **without needing a host** and spreads by itself across a network is a **worm** (see Part C) — that distinction is a classic MCQ.

The life cycle — the numbered answer

A virus goes through recognisable phases. This numbered list is the ideal shape for a 5-mark "how does a virus work" answer:

1. **Infection / insertion.** The virus code is attached to a host (program, boot sector, or document macro). This usually happens when you run infected software, open an infected attachment, or boot from an infected disk / USB.
2. **Dormant phase (latency).** The virus sits idle, doing nothing visible, waiting for a trigger. Not all viruses have this phase.
3. **Trigger (activation).** A condition fires it — a particular date (e.g. the "Friday the 13th" virus), a number of executions, or a specific user action.
4. **Replication / propagation.** Each time the infected host runs, the virus copies itself into other clean files or disks, spreading the infection.
5. **Execution / payload / damage.** The virus delivers its **payload** — the harmful action: deleting files, corrupting data, displaying messages, slowing the system, or opening a backdoor. Some payloads are merely annoying; some are destructive.

Exam phrasing tip. Examiners love the words "**replicate**" and "**payload**". Use both in any virus answer. Note: replication is *always* present; a destructive payload is *not* — some viruses only replicate and consume resources.

Key points

- A virus is **man-made software**, written deliberately. (The 2024 paper asked "Why do people write these?" with the intended answer being about **viruses** — motives include mischief, vandalism, showing off skill, financial gain, and disruption.)
- A virus **cannot infect hardware directly** — it corrupts *data and programs*. (2024 MCQ: "A computer virus is unable to do —" answer "**Disabling hardware**". Modern nuance: malware *can* damage hardware indirectly, e.g. overwriting firmware/BIOS or overheating — mark this as $\triangle$ verify against the intended textbook answer, which says a virus cannot physically destroy hardware.)
- Viruses spread through **executable files, boot sectors, email attachments, infected removable media (USB/CD), macros in documents, and network shares**.
- **Plain-text and pure-data formats generally cannot carry a virus** because they contain no executable code. (2024 MCQ: "Which document format cannot contain a virus?" — the intended answer was **.rtf** $\triangle$ verify. The safe underlying principle: formats that support **macros or embedded code** (.doc, .docx, .xls, .pdf with scripts) *can* carry malware; a format with no code-execution capability cannot. **.txt** is the cleanest true example.)

B. Types of viruses — the syllabus's own taxonomy (HIGH PRIORITY)

This is the single most examinable sub-topic in the unit, and it is worded in **1990s vocabulary you must reproduce exactly**. Learn these six categories in the syllabus's own words.

#	Virus type	What it infects / how it works	Key identifier
1	Command-processor infection	Infects the command interpreter — <code>COMMAND.COM</code> in DOS (the shell that reads and executes your commands). Because the command processor is loaded and run constantly, the virus gains control very frequently.	Targets the shell / <code>COMMAND.COM</code>
2	Boot-sector infection	Infects the boot sector (the Master Boot Record / DOS boot record) of a disk. It loads into memory before the OS , every time you boot from the infected disk. Spreads mainly via infected floppies/USB left in the drive at boot.	Runs at boot, before the OS
3	Executable-file infection	Attaches to executable program files (<code>.EXE</code> , <code>.COM</code> , <code>.SYS</code>). When you run the program, the virus runs first (or alongside) and infects other executables. The classic "file virus" / "parasitic virus".	Rides on <code>.EXE/.COM</code> programs
4	File-specific infection	Targets one particular named file or one particular type of file — it is selective, only infecting a specific target rather than any executable.	Highly selective — one file/type
5	Memory-resident infection	Loads itself into RAM and stays resident (TSR — Terminate-and-Stay-Resident) . It stays active in memory even after the host program ends, and infects every program that runs or every disk that is accessed while it is resident.	Lives in RAM, infects on-the-fly
6	Macro virus	Written in the macro language of an application (e.g. VBA in MS Word/Excel). It infects documents and templates (e.g. Word's <code>Normal.dot</code>), not programs. Spreads when infected documents are shared. Was the most common virus type in the late 1990s because documents are shared far more than programs.	Infects documents via macros

How to keep them straight (the memory hook)

Group them by **what they attach to**:

- **The shell** → command-processor virus.
- **The disk's boot area** → boot-sector virus.
- **Program files** → executable-file virus (any) and file-specific (one specific target).
- **Where it lives while active** → memory-resident (stays in RAM) vs non-resident (runs, infects, exits).
- **Documents, not programs** → macro virus.

A useful extra distinction (often asked): a **resident** virus stays in memory and works in the background; a **non-resident / direct-action** virus executes, finds and infects a few targets, then hands control back and exits without staying in memory. Also worth naming: **polymorphic virus** (changes its own code/signature on each infection to evade scanners) and **stealth virus** (actively hides its presence from antivirus, e.g. by intercepting disk reads). These two are not in the six-type list but are common MCQ distractors.

MCQ traps for virus types

- **Boot-sector virus loads before the OS** — a favourite "true/false" fact. True.
- **Macro virus infects documents, not programs** — do not say it infects `.EXE` files.

- A **memory-resident** virus is **not** the same as a boot virus; a boot virus *becomes* resident, but resident refers specifically to staying in RAM.
- **Command-processor virus** = **COMMAND.COM**, not "any command you type".

C. Worms, trojans and other malware — for context and contrast

The syllabus's virus unit sits inside a broader malware picture. You need the contrasts because "differentiate virus and worm" is a stock 5-marker and the distinctions are prime MCQ material.

Malware	Self-replicates?	Needs a host?	Spreads by	One-line definition
Virus	Yes	Yes (attaches to a file/program/boot sector)	Running the infected host; sharing files/media	Malicious code that attaches to a host and replicates when the host runs
Worm	Yes	No (standalone)	By itself, across networks (exploits, email)	A standalone program that self-replicates and spreads over networks without a host
Trojan horse	No	Disguises itself as legitimate software	User is tricked into installing it	Malware disguised as a useful program; does NOT self-replicate
Ransomware	Sometimes	Often delivered by trojan/worm	Phishing, exploits	Encrypts the victim's files and demands payment for the key
Spyware	No	—	Bundled with software, drive-by	Secretly collects user information
Backdoor / RAT	No	—	Trojan delivery	Lets an attacker take remote control of the PC over the internet

Notes that turn into MCQs:

- **The virus-vs-worm line = self-propagation.** A worm spreads **on its own** across a network; a virus needs the user to run/open the host. (The syllabus itself often lumps these, but the exam distinguishes them.)
- **Trojan does not replicate** — that is its defining contrast with viruses and worms. (2024 MCQ: "a program that allows someone to take control of another user's PC via the internet" → **Backdoor Trojan**.)
- **Virus hoax** = a fake warning (usually a chain email) telling you a non-existent virus will destroy your PC and urging you to forward the message. It carries no code — the "damage" is the panic and wasted effort. (2024 MCQ: "Which one is a virus hoax?" → the famous **"Good Times"** hoax.)
- **Spam / UCE / junk mail** are the same thing (Unsolicited Commercial Email) — a 2024 MCQ grouped Spam / Junk Mail / UCE together and asked which was the odd one out (Gmail).

D. Protecting, recognising, and dealing with an infection

These three are natural sub-parts of a 10-mark or 15-mark question ("Discuss the types of viruses, how to recognise an infection and how to deal with it"). Keep each as a numbered list.

D.1 Protecting the PC (prevention)

1. **Install and update antivirus / anti-malware software**, and keep its virus definitions current — an out-of-date scanner is nearly useless.
2. **Keep the OS and applications patched** — most infections exploit known, already-fixed vulnerabilities.
3. **Use a firewall** (software and/or hardware router) to block unauthorised network access.
4. **Do not run unknown executables or open unexpected email attachments**; be wary of links.
5. **Scan removable media (USB, CD/DVD) before use**; disable autorun.
6. **Disable macros by default** in Office; enable only for trusted documents.
7. **Take regular backups** kept offline/separate — the only reliable recovery from destructive malware and ransomware.
8. **Use standard (non-administrator) user accounts** for daily work to limit what malware can do.
9. **Download software only from trusted / official sources.**
10. **Configure the browser for security** and keep a **separate network for internet-facing machines** where possible. (Both appear as 2024 MCQ "safety tip" options; the *wrong* option — i.e. NOT a safety tip — was "always explore/open money-related mails".)

D.2 Recognising an infection (symptoms)

Common signs that a PC may be infected:

1. The computer runs **noticeably slower** than usual, or freezes/crashes often.
2. Programs **fail to start, behave oddly, or close unexpectedly**.
3. **Unusual error messages, pop-ups or on-screen messages** appear.
4. **Files or folders disappear, get renamed, change size, or become corrupted**.
5. **Free disk space or memory shrinks** for no clear reason (the virus is replicating).
6. The system **boots slowly or fails to boot**; the OS won't load (2024 MCQ hint: "PC boots but will not start the OS" as a troubleshooting scenario — often the boot area).
7. **Antivirus software is disabled** or cannot be updated (malware often kills it).
8. **Unexpected network activity**, the modem/disk light active when idle, or programs accessing the internet on their own.
9. **Emails you did not send** go out from your account.

D.3 Dealing with an infection (removal / recovery)

1. **Disconnect from the network / internet** to stop it spreading and to cut off any backdoor.
2. **Do not panic and do not reboot repeatedly** — some viruses do damage on each boot.
3. **Run a full scan with up-to-date antivirus**; let it **quarantine or clean** the infected files.
4. If the running OS is compromised, **boot from a clean rescue disk / bootable antivirus media** and scan from outside the infected OS.
5. **Delete or repair** the files the scanner cannot clean; for boot-sector infections, repair the boot record (e.g. **FDISK /MBR** in DOS-era systems △ verify for the specific OS).
6. **Restore lost/corrupted data from a clean backup.**
7. **Change passwords** (they may have been captured) after the system is clean.
8. In severe cases, **reformat and reinstall the OS** from trusted media, then restore data.
9. **Patch the vulnerability** that let it in, so you are not re-infected immediately.

Likely exam questions — Viruses

5-mark

- What is a computer virus? Explain how it works. (*definition + numbered life cycle*)
- Differentiate between a virus and a worm. (*two-column table*)
- What is a Trojan horse? How does it differ from a virus? (*table*)
- List any five symptoms that indicate a computer may be infected.
- What is a macro virus? Why did it become so common?
- What is a virus hoax?

10-mark

- Explain the different types of computer viruses. (*the six-row table — this is the safe banker; add polymorphic/stealth for extra marks.*)
- How can a PC be protected from viruses? Also explain how to recognise an infection. (*two numbered lists.*)
- Compare virus, worm, trojan and ransomware. (*four-row table + one line each.*)

15-mark

- Discuss computer viruses in detail: what they are, how they work, their types, how to recognise an infection and how to deal with one. (*This is the "everything" question — structure it exactly as Parts A→D above: definition + life cycle, the six types table, recognition list, protection + removal lists. A labelled structure with tables outscores prose easily.*)

MCQ traps — Viruses (from the real 2024 paper and their kin)

- "A virus is unable to do —" → **disabling hardware** (per textbook answer). △ verify.
- "Where is your office **not** vulnerable?" → **SMPS** (the power supply is not an infection vector; internet/email/CD are).
- "Why do people write these?" → **Virus**.
- "Which is a virus hoax?" → **Good Times**.
- "Take control of another user's PC via the internet" → **Backdoor Trojan**.
- "Which document format cannot contain a virus?" → intended **.rtf** (△ verify; the sound principle is: a format with no macros/embedded code cannot carry one — **.txt** is the cleanest case).
- "Which is NOT a safety tip while using the internet?" → **"Always explore money-related mails."**

PART 2 — SYSTEM ANALYSIS & DESIGN (Paper-II, Unit 4)

Shared-core pointer. The software-process side of this unit — waterfall / prototype / spiral / RAD models, cost estimation, testing strategies, quality assurance — is treated in depth in the shared-core software-engineering notes (conventionally **CORE_07_SoftwareEngineering.md** — △ confirm the exact filename by listing `Materials\_Shared\`; at time of writing the `_Shared` folder contains `CORE_01_DigitalLogic`, `CORE_02_ComputerOrganization`, `CORE_03_OperatingSystems`, `CORE_04_UnixLinux`, `CORE_05_ComputerNetworks`). **This file covers the diploma-specific angle:** the *systems-theory vocabulary* (system concept, characteristics, elements, and the classification of systems), *information-system types*, the *SDLC phases at diploma depth*, the *role of the system analyst*, and *failure factors* — the material a B.Tech course usually skips because it is MIS vocabulary rather than software engineering.

E. The system concept — definition, characteristics, elements

F. Types of system (HIGH PRIORITY — five classifications)

The syllabus explicitly names **physical/abstract** and **open/closed**; the standard SAD textbook adds three more pairs. Each pair is a clean two-column table — pre-build all five.

1. Physical vs Abstract

Physical system	Abstract system
Tangible entities that physically exist	Conceptual / non-physical — ideas, models, formulas
Can be static (fixed) or dynamic (changing)	Exists as a representation of a real system
Example: a computer, a car, a payroll office	Example: a mathematical model, a set of formulas, a design blueprint

2. Open vs Closed

Open system	Closed system
Interacts freely with its environment — takes input and returns output	Cut off from its environment; does not interact with it
Adapts to environmental change	Isolated, self-contained; largely theoretical (truly closed systems are rare)
Example: a business organisation, a living organism	Example: a sealed chemical reaction; a fully isolated program

3. Deterministic vs Probabilistic

Deterministic system	Probabilistic (stochastic) system
Behaviour is fully predictable ; output is known if input and state are known	Behaviour involves uncertainty / probability ; output cannot be predicted with certainty
Example: a computer program, a correct arithmetic operation	Example: weather forecasting, inventory demand, a warehouse system

4. Man-made vs Natural (*commonly listed*)

Man-made system	Natural system
Designed and built by humans	Exists in nature, not created by humans
Example: a transport system, an information system	Example: the solar system, an ecosystem

5. Permanent vs Temporary (*commonly listed*)

Permanent system	Temporary system
Exists for a long, indefinite period	Set up for a limited time then dismantled
Example: an organisation's accounting system	Example: a system built for a one-off event/project

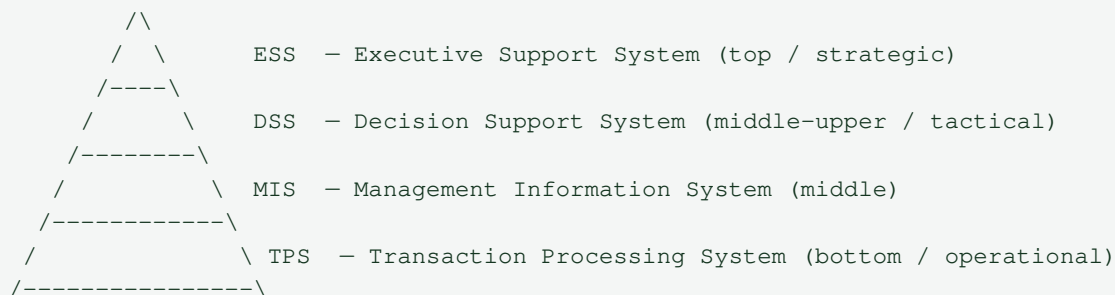
MCQ calibration (2024 matching question): open = "interacts freely with its environment"; closed = "cut off from its environment"; physical = "tangible entities, static or dynamic"; abstract = "non-physical / conceptual, model of a real system."
Learn those four phrases word-for-word — the matching question used them almost verbatim.

G. Information systems (TPS / MIS / DSS / ESS)

Concept

An **information system** is a man-made system that **collects, processes, stores and distributes information** to support the operations, management and decision-making of an organisation. Its basic elements are **people, hardware, software, data and procedures** (sometimes networks is added as a sixth). (2024 MCQ: "basic element of an information system?" → People / Hardware / Software → All of the above.)

Information systems are usually drawn as a **pyramid**, matching the levels of management they serve — bottom = operational/day-to-day, top = strategic:



Type	Full form	Serves	What it does	Example
TPS	Transaction Processing System	Operational staff	Captures, classifies, stores, updates and retrieves day-to-day transaction data ; the base that feeds all other systems	Billing, payroll, order entry, ATM transactions
MIS	Management Information System	Middle management	Summarises TPS data into routine reports for monitoring and control	Monthly sales report, inventory summary
DSS	Decision Support System	Middle/upper management	Interactive tools + models to support semi-structured decisions ("what-if" analysis)	Sales forecasting, budgeting model
ESS / EIS	Executive Support / Information System	Top executives	Highly summarised, big-picture information for strategic decisions	Company-wide performance dashboard
(OAS)	Office Automation System	All office staff	Supports communication/productivity (email, word processing, scheduling)	MS Office, email, calendars
(KWS)	Knowledge Work System	Knowledge workers	Supports creation of new knowledge (CAD, analysis tools)	CAD workstations

MCQ calibration (2024): "Which is **not** a Computer-Based Information System (CBIS)?"

→ options TPS, MIS, **Networking System**, OAS → answer **Networking System**. And "which system captures, classifies, stores, maintains, updates and retrieves data for record keeping and input to other CBIS?" → **TPS**. And "which is **not** part of OAS?" → **System designing** (mailing, typing, scheduling are OAS; system design is not).

H. The System Development Life Cycle (SDLC) — phases (HIGH PRIORITY)

Concept

The **SDLC** is the structured sequence of stages a system passes through from first idea to final retirement. Different books split it into 5–8 phases with slightly different names; the **substance is always the same**. Learn one canonical ordered list and be able to state the **deliverable (output) of each phase** — examiners reward that.

The phases (canonical seven-phase version)

#	Phase	What happens	Deliverable / output
1	Preliminary investigation / Problem identification	Recognise the problem or need; do a quick scope check.	Problem statement; project request
2	Feasibility study	Is the project worth doing? Assess technical, economic (cost–benefit), operational, legal and schedule feasibility.	Feasibility report; go / no-go decision
3	Requirements analysis (System analysis)	Study the current system; gather and analyse user requirements; model data & processes (DFDs, ER).	SRS (System/Software Requirements Specification)
4	System design	Decide how the system will meet the requirements — structure of files, databases, inputs, outputs, processes and screens/interfaces.	Design document (logical + physical design)
5	Coding / development (implementation)	Programmers write and unit-test the code from the design.	Working program modules
6	Testing	Verify the system against requirements — unit, integration, system, acceptance testing.	Tested, validated system; test reports
7	Implementation / deployment & Maintenance	Install, convert data, train users, go live; then maintain — fix bugs and enhance over the system's life.	Operational live system; maintenance updates

Two-way memory aid: "*Investigate → Feasibility → Analyse → Design → Code → Test → Implement & Maintain.*" (2024 MCQ: "first phase of SDLC?" → **Problem Analysis** among the options given; "phase where structure of files, databases, input, output, processes and screens is decided?" → **Designing**; "which is NOT a phase of SDLC?" → **Networking**.)

Note on models. The *phases* above are the life cycle; the *models* (Waterfall, Prototyping, Spiral, RAD/DSDM) are different ways of *sequencing/iterating* those phases — covered in the shared-core SE notes. Two diploma-level facts worth carrying here:

RAD (Rapid Application Development) is also known as / associated with **Prototyping and DSDM** (2024 MCQ), and **feasibility study types** are **technical, economic, operational, legal and schedule** (2024 MCQ: "type of feasibility?" → All of the above).

I. Role of the system analyst; factors causing failure

Role of the system analyst

The **system analyst** is the **bridge between the users/management and the technical developers** — the person who understands the business problem *and* enough technology to specify a solution. Present as a numbered list of roles/responsibilities:

1. **Investigator / fact-finder** — studies the existing system, gathers requirements (interviews, questionnaires, observation, document study).

2. **Problem-solver / analyst** — analyses requirements and defines what the new system must do; models data and processes.
3. **Communicator / liaison** — communicates effectively, one-on-one, with users, business managers **and** programmers; translates business needs into technical terms and back.
4. **System designer** — specifies the design of files, databases, inputs, outputs and interfaces.
5. **Project manager / coordinator** — plans, schedules and coordinates the project team and resources.
6. **Change agent** — introduces and manages the change the new system brings; trains and supports users.
7. **Ethical professional** — deals **fairly, honestly and ethically** with team members, managers and users, and **maintains confidentiality**.

Common role labels an analyst wears (2024 MCQ): **business analyst, requirements analyst, project manager** — the question asked which is NOT a role and the answer was "All of the above" *are* roles. Also (2024): "not important for analysts to maintain confidence" is a **false** statement — confidentiality *is* required.

Factors causing failure in the SDLC / a system project

A stock 10-marker. Why systems fail:

1. **Incomplete, unclear or changing requirements** — the biggest single cause.
2. **Poor communication** between users, analysts and developers.
3. **Lack of user involvement / management support.**
4. **Unrealistic schedule, budget or scope** (over-optimistic planning).
5. **Inadequate feasibility study** — building the wrong thing.
6. **Poor project management** and lack of proper planning/control.
7. **Insufficient testing** — defects reach production.
8. **Resistance to change** / inadequate user training.
9. **Technology risk** — chosen technology is immature or unsuitable.
10. **Inadequate maintenance / no post-implementation review.**

Likely exam questions — System Analysis & Design

5-mark

- Define a system. State its characteristics.
- Explain the basic elements of a system with a diagram. (*IPOFC + feedback*)
- Differentiate between an open system and a closed system. (*table*)
- Differentiate between a physical and an abstract system. (*table*)
- What is a deterministic system? Give an example.
- Differentiate between TPS and MIS.

10-mark

- Explain the different types of systems with examples. (*the five paired tables — a strong banker.*)
- Describe the types of information systems (TPS, MIS, DSS, ESS) with a pyramid and examples.
- Explain the role of the system analyst.
- Discuss the factors that cause failure in system development.

15-mark

- Explain the SDLC in detail — describe each phase and its deliverable. (*the seven-phase table; add a short line on Waterfall vs Prototyping vs Spiral for extra depth.*)
- "A system analyst is the bridge between users and developers." Discuss the role of the

system analyst and the factors that cause a system project to fail.

MCQ traps — System Analysis & Design

- Basic elements of a system → Input, Process, Output, Control, Feedback (**not** Procurement).
- CBIS types → TPS, MIS, DSS, ESS, OAS (**not** "Networking System").
- OAS includes mailing, typing, scheduling (**not** "system designing").
- SDLC phases (**not** "Networking").
- RAD ⇔ Prototyping / DSDM.
- Feasibility types → technical, economic, operational, legal, schedule.
- Structured tools: DFD, ERD, STD are structured tools; **RAD is a methodology, not a tool** (2024 MCQ: "which is NOT a structured tool?" → **RAD**).
- Charles Bachman won the Turing Award for **DBMS** (network data model) — a 2024 SAD-section MCQ, easy to confuse; note it is DBMS, not SAD.

PART 3 — INTERNET TECHNOLOGY (Paper-I, Unit 6)

Shared-core pointer. The *network fundamentals* underneath the internet — LAN/MAN/WAN, the OSI and TCP/IP models, transmission media and modes, topologies, and connectivity devices (hub/switch/router/bridge/gateway) — are covered in depth in **Materials_Shared\CORE_05_ComputerNetworks.md** (and the Paper-I networks unit, §5). **This part covers the diploma-specific internet layer:** intranet vs internet, the internet *services*, the WWW / browsers / search engines, and the *connectivity types* (dial-up / leased / ISDN / VSAT) — which are dated, specific, and heavily examined.

J. Intranet vs Internet vs Extranet

Concept

- **Internet** — the global, public "network of networks" connecting millions of computers worldwide; open to everyone. (2024 MCQ: "Once restricted to a few scientists, it quickly became universal. It reaches everywhere." → *Internet*.)
- **Intranet** — a **private network internal to one organisation**, using the same internet technologies (TCP/IP, browsers, web pages) but **accessible only to authorised members** of that organisation.
- **Extranet** — a **controlled extension of an intranet to selected outsiders** (suppliers, partners, customers) over the internet, usually secured by login/VPN.

Feature	Intranet	Internet	Extranet
Scope	Within one organisation	Global / worldwide	Organisation + selected partners
Access	Private, authorised users only	Public, anyone	Restricted external users
Ownership	One organisation	No single owner	One org + its partners
Security	High (behind firewall)	Low / open	Controlled (login/VPN)
Example use	Company noticeboard, HR portal	Public websites, email	Supplier order portal, B2B

MCQ calibration (2024): "Which service is **not** possible with an intranet?" →

Public Web sites (an intranet is private; public sites belong to the internet.
Private discussion groups, access to legacy databases and teleconferencing are possible internally).

K. Internet services

The syllabus names these services explicitly — learn one line for each:

Service	What it does
Email (Electronic mail)	Send/receive messages and attachments. Sending uses SMTP ; retrieving/storing in the mailbox uses POP3 or IMAP .
Chatting / Instant messaging	Real-time text (and voice/video) conversation between users; chat rooms use an avatar/nickname as your online identity; often implemented in Java △ (2024 MCQ said Java).
Bulletin Board System (BBS) / Newsgroups (Usenet)	Post-and-read message boards on topics; Usenet is one of the oldest internet services, older than the WWW. A web forum is the modern alternative to a real-time chat room.
Video conferencing	Real-time audio + video meeting between remote participants. Components: camera, microphone, monitor/display, speakers, codec, network — (2024 MCQ: an "interpreter" is NOT a component). Apps: NetMeeting, CU-SeeMe, WebEx.
FTP (File Transfer Protocol)	Transfer arbitrary files between computers over the internet (upload/download). (2024 MCQ: "transfer arbitrary files across the internet" → FTP.)
Telnet	Log in to and use a remote computer over the network as if sitting at it (a remote terminal). (Note: Telnet is a service/protocol, NOT a type of internet connection — a 2024 trap.)
WWW (World Wide Web)	The system of interlinked hypertext documents (web pages) accessed via browsers over the internet.
VoIP	Voice calls over the internet (telephone-to-telephone, or PC-to-PC), e.g. Skype.
VoD / Podcast / Streaming	On-demand audio/video; a playback buffer allows smooth streaming despite network jitter.

Netiquette & terminology (all real 2024 MCQs): **Netiquette** = suggested rules for polite online behaviour; **Internauts / Netizens** = internet users; **postmaster@domain** = the address to contact when you know a domain but not a username; **mailbot** = an automatic email auto-responder; **user agents** = email client programs (Gmail, Outlook, Thunderbird, Apple Mail).

L. The World Wide Web, browsers and search engines

WWW essentials

- **WWW** = a collection of **hypertext/hypermedia documents (web pages)** linked by **hyperlinks** and served over the internet. Invented by **Tim Berners-Lee in 1989** (2024 MCQ).
- Its enabling technologies: **HTML** (page markup), **HTTP** (transfer protocol), and the **URL** (address).
- **URL (Uniform Resource Locator)** — the unique **address of a file/resource on the internet** (2024 MCQ). Form: `protocol://host/path` e.g. `https://www.example.com/page.html`.

- **W3C (World Wide Web Consortium)** develops the standards/guidelines for the long-term growth of the Web (2024 MCQ).
- **Web server** — software/computer that **stores, retrieves and distributes the Web's files** (serves pages to browsers) (2024 MCQ). **Web browser** — the client program that requests and displays those pages.

Web browser

A **browser** is the client application used to access and display web pages (Chrome, Firefox, Edge, Safari). Typical **elements of the browser window**: **title bar, menu bar, toolbar, address/URL bar, viewing/content window, status bar, scroll bars, tabs, bookmarks/favourites bar**. (2024 MCQ asked which is NOT an element of the browser window.)

Search engines — how searching works

A **search engine** (Google, Bing) is a service that helps users **find information on the Web by keyword**. Its core requirement is a **huge database/index** of web pages (2024 MCQ: "most required to a search engine" → **Database**). It works in three stages:

1. **Crawling** — automated programs (**spiders / crawlers / bots**) follow links and fetch web pages.
2. **Indexing** — the fetched pages are analysed and stored in a searchable **index** (by keywords).
3. **Searching / ranking** — when you enter a query, the engine matches it against the index and returns results **ranked by relevance** (using text matching, keyword frequency, link popularity, etc.). (2024 MCQ: "approach NOT used in web searching" → *Priority setting was the odd option; text matching, indexing, pattern matching, keyword frequency are used.*)

M. Connectivity — ISP and connection types (HIGH PRIORITY)

Concept

To reach the internet you connect through an **ISP (Internet Service Provider)** — a company that sells internet access (and often email, hosting, etc.). You need: a **device + a modem/router + a physical line + an account with an ISP**. The *type* of line/account is the examinable part, because the options are dated and specific.

Connection-type comparison table (learn this cold)

Connection	Line / medium	Speed (approx)	How it works	Notes
Dial-up	Ordinary telephone line + modem	~ 56 kbps (very slow)	Modem dials the ISP over the voice phone line; ties up the phone while connected	Oldest, cheapest, obsolete; connection established using a modem
Leased line	Permanent dedicated telephone line	High, fixed (e.g. T1 ≈ 1.544 Mbps △ verify)	A private line permanently connected to the ISP — always on, dedicated bandwidth	Expensive; for businesses; created with a permanent telephone line

ISDN	Digital telephone line (Integrated Services Digital Network)	64 / 128 kbps (BRI: 2B+D) △ verify	Sends digital voice+data over the phone network; faster and cleaner than dial-up	Narrowband ISDN and Broadband ISDN (B-ISDN) exist
Cable	Coaxial cable TV network	Several Mbps+	Shares the cable-TV line; a cable modem connects the PC	Faster than dial-up; shared bandwidth in a locality
DSL / Broadband	Telephone copper (DSL) or fibre	Mbps–Gbps	"Always-on" high-speed digital line; DSL runs data above voice frequencies so phone stays free	The modern default
VSAT	Satellite (Very Small Aperture Terminal)	Varies	An earthbound station with a small dish communicates via a satellite — used where cabling is impractical (remote areas)	Satellite communication; higher latency
Wireless / Wi-Fi / mobile	Radio (Wi-Fi, 3G/4G/5G)	Varies	Connect via radio frequency to an access point / cell tower	Hotspot = a Wi-Fi access point

MCQ calibration (2024 matching question): VSAT = "an earthbound station used in satellite communication"; Dial-up = "established using a modem"; Cable connection = "created/carried over cable"; Leased connection = "created with a permanent telephone line." Learn those four phrases verbatim.

Also (2024): Telnet is **NOT a type of internet connection** (it is a remote-login service); Bluetooth is **NOT a way to connect to the internet** in the intended answer (Wi-Fi, hotspot are) △ verify — Bluetooth tethering technically exists, but the exam treated Bluetooth as the odd one out.

Likely exam questions — Internet Technology

5-mark

- Differentiate between the internet and an intranet. (*table*)
- What is a URL? Give its structure with an example.
- Differentiate between a web browser and a web server.
- List and briefly explain any five internet services.
- What is video conferencing? Name its components.
- What is an ISP?

10-mark

- Explain the various services provided by the internet. (*the services table*.)
- Explain the different types of internet connections. (*the connectivity table — a strong banker; VSAT/ISDN/leased/dial-up are the diploma-favourite content*.)
- How does a search engine work? Explain crawling, indexing and ranking.
- Differentiate between intranet, internet and extranet. (*three-column table*.)

15-mark

- Explain internet technology in detail: intranet vs internet, the main internet services, the WWW with browsers and search engines, and the types of internet connectivity. (*The "everything" question — structure exactly as Parts J→M*.)

MCQ traps — Internet Technology

- Intranet **cannot** host **public web sites** (it is private).

- **SMTP** sends mail; **POP3** receives/stores it in the mailbox.
- **Usenet** is one of the **oldest** internet services (older than the WWW).
- **FTP** transfers arbitrary files; **Telnet** is remote login (NOT a connection type).
- **Video conferencing** components do **not** include an "interpreter".
- Search engine's most-required asset = **database/index**; "priority setting" is not a search approach.
- **URL** = address of a file on the internet; **W3C** sets Web standards; **Tim Berners-Lee** invented the WWW (1989).
- Connection matching: **VSAT** = **satellite** / **earthbound station**, **dial-up** = **modem**, **leased** = **permanent line**.

One-paragraph summary, if you read nothing else

Three cheap-mark units. For **viruses**, learn the syllabus's own six-type taxonomy (command-processor, boot-sector, executable-file, file-specific, memory-resident, macro), the virus-vs-worm-vs-trojan contrasts, and the three numbered lists (protect / recognise / deal-with). For **system analysis**, this is MIS vocabulary, not software engineering: define a system and its IPOFC elements, classify systems five ways (physical/abstract, open/closed, deterministic/probabilistic, man-made/natural, permanent/temporary), name TPS/MIS/DSS/ESS, recite the SDLC phases with their deliverables, and know the analyst's role and the failure factors. For **internet technology**, know intranet-vs-internet, the internet services, the WWW/browser/search-engine story, and above all the dated connectivity table (dial-up, leased, ISDN, VSAT). All three are more about reproducing the textbook's exact words than about understanding — so drill the phrasing, build every "differentiate" table in advance, and answer with tables and numbered lists.

Where the rest of these units live

You need...	Look in
Network fundamentals (OSI, TCP/IP, media, topologies, devices)	Materials_Shared\CORE_05_ComputerNetworks.md + Paper-I §5
Software process models, cost estimation, testing, QA	CORE_07_SoftwareEngineering.md △ confirm filename in _Shared\
Legacy protocols (NetBEUI, IPX/SPX), IEEE 802.x	CORE_05_ComputerNetworks.md
Hardware, storage, motherboard	CSDip_01_HardwareAndDevices.md
MS Office / MS-DOS / Windows	CSDip_02_MSOoffice_and_DOS.md (written by another agent)
Exam dates, strategy, day-by-day plan	CSDip_00_OVERVIEW.md